

Checking whether SDSS has a spectrum for me

Learning goals

Through this tutorial, you will learn:

- How to use the "allspec" file either locally or on SciServer to check whether an object you are interested in has SI
- How to track down the relevant SDSS-V BOSS (optical) parameters and spectrum for galaxies and quasars.
- How to track down the relevant SDSS-V BOSS (optical) and APOGEE (infrared) parameters and spectrum for stars.

Introduction

SDSS has observed spectra in several ways over the past 20+ years. Specifically, we have used the SDSS and BOSS opt modes, and we have used the two APOGEE spectrographs in single-fiber mode. Meanwhile, this data has been processed and is processed into both individual visit and coadded spectra.

To help users track down the relevant information for objects of interest, SDSS has created the `allspec` file. This file lists all visit spectra and all coadds from every spectrograph. In this file, every object may appear once or more, if they have multiple visits.

Objects in SDSS are tracked through the `sdss_id` identifier. This is designed to correspond to unique objects on the sky, the "same object" across different spectra.

Imports

We will need several Python modules for our tasks:

- `os` because we usually need it
- `matplotlib.pyplot` for plotting
- `matplotlib` because I'm persnickety about TeXing labels

- `numpy` because we have to do things with numbers and arrays
- `astropy.io.fits` to read spectra files
- `fitsio` for efficiently reading very large files (`spAll`, `specObjAll`)
- `astropy.coordinates` for spatial matching
- `sdss_access` to track down files
- `polars` for reading in the very large `allspec` file
- `scipy` part of `astropy[all]`, otherwise needed on it's own for coordinate matching

Caveats: If you are running this notebook on your own machine, there are few things you should make note of:

1. Some of the files loaded in this notebook are very large and use a lot of RAM (seriously: > 32 GB RAM is required).
2. If you don't have a properly configured Tex distrubution, you may need to comment out the line `matplotlib.rcF` cell.
3. The first time you check for a local file on your "local" "SAS", the filename may be created as `.fits` whereas the c cell fixes the problem.

A note about parquet files

For DR20, a parquet version of the `allspec` file has been made available. Using the parquet file results in reading the f as much memory compared to the `fits` file. These efficiency gains make parquet files an easy choice. There are a few m

SDSS File Access

`sdss_access` provides ways to track down SDSS files in the data structure.

This notebook is perhaps easiest to run on SciServer. But if you are working on a machine without a full copy of the S to download and cache files locally. See <https://sdss-access.readthedocs.io> for more information!

Finding and Reading the `allspec` file

`sdss_path` gives us an easy way to find the path to the `allspec` file.

Allspec file path: /data/sdss/dr20/spectro/allspec/1.0.2/allspec-dr20-1.0.2.parquet
File exists locally: True

Successfully loaded allspec dataset shape: (27671504, 38)
First 5 rows preview:
shape: (5, 38)

allspec_id --- str	multiplex _id --- str	sdss_phase --- i64	observatory --- str	...	has_mwmstar --- bool	astraversions --- list[str]	manga_id --- str	specobj_id --- decimal[38,0]
sdss3-apo-sdss-103-1960-53289-...	sdss3-apo-sdss-103-1960-53289	3	APO	...	false	null	null	2206764147469818880
sdss3-apo-sdss-103-1960-53289-...	sdss3-apo-sdss-103-1960-53289	3	APO	...	false	null	null	2206764422347725824
sdss3-apo-sdss-103-1960-53289-...	sdss3-apo-sdss-103-1960-53289	3	APO	...	false	null	null	2206764697225632768
sdss3-apo-sdss-103-1960-53289-...	sdss3-apo-sdss-103-1960-53289	3	APO	...	false	null	null	2206764972103539712
sdss3-apo-sdss-103-1960-53289-...	sdss3-apo-sdss-103-1960-53289	3	APO	...	false	null	null	2206765246981446656

Matching by position to allspec

Let's say we have some position of interest. Let's figure out if we have SDSS spectra of any sort near it!

I like the location (ra, dec) = (177.78, -0.73), just because I do. Let's put this in the `astropy.coordinates.SkyCoord` o

Each `sdss_id` has a specific RA and Dec, so we only need to match to the unique set of `sdss_ids`. We will put those c

Now it is easy to match. Let's look at the closest one to our desired position.

```
Match index in unique list (iminsep): 3027677
Matched row index in allspec (match_indx): 9508842
Matched SDSS ID: 74807581
Separation (arcsec): 0.3005637335987949
```

Information in allspec

Now that we have found a spectrum, we can take a look at the information in `allspec`. Below you can see several typ

- The `allspec` and `multiplex` identifiers (just unique identifiers)
- Information about the observatory, SDSS phase, and instrument
- The `sdss_id` and `catalogid` of the target object.
- Many columns of spectroscopic IDs. Only one set of identifiers will be filled with meaningful information.
- The name and location of the spectrum file (useful!) and a link to SkyServer

```
allspec_id :: sdss5-apo-apogee-1.3-4054-59657-185
multiplex_id :: sdss5-apo-apogee-1.3-4054-59657
sdss_phase :: 5
observatory :: APO
instrument :: apogee
sdss_id :: 74807581
catalogid :: None
fiberid :: 185
ifudsgn :: None
plate :: 4054
fps_field :: None
plate_or_fps_field :: None
mjd :: 59657
run2d :: None
run1d :: None
coadd :: None
```

```

apred_vers :: 1.3
drpver :: None
version :: 1.3
programname :: mwm_dust_core
survey :: SDSS-V
sas_file :: apVisit-1.3-apo25m-4054-59657-185.fits
cas_url :: https://skyserver.sdss.org/public/VisualTools/explore/summary?apid=2M06533798-0119171
sas_url :: https://data.sdss.org/sas/dr19/spectro/apogee/redux/1.3/visit/apo25m/103442/4054/59657/apVi:
ra :: 103.408257219
dec :: -1.32142421849
healpix :: 100755
healpixgrp :: 100
apogee_id :: 2M06533798-0119171
apogee_field :: 103442
telescope :: apo25m
file_spec :: apVisit
apstar_id :: apogee.apo25m.stars.2M06533798-0119171
visit_id :: apogee.apo25m.1.3.4054.59657.185
has_mwmstar :: True
astra_versions :: shape: (2,)
Series: '' [str]
[
    "0.6.0"
    "0.8.1"
]
mangaid :: None
specobjid :: None

```

Finding all spectra

We can also find all of the spectra associated with the `sdss_id`. This is useful for objects with repeat spectroscopy.

```

Matched spectrum indices array (iallmatch): [ 9508842  9560683  9589897  9652277 10278507 11950596 125:
 18367087 18507005 18889701 19470954 19476116 21386477 26418544 26560828
 26936607]
Total matching spectra count: 17

```

We can look at what these spectra actually are. It can take some getting used to to figure out this information. What tl

- There are three distinct BOSS spectra, which were taken during SDSS-V.
- We can see that for each MJD (59654 through 59656) there is both a "daily" and an "epoch" coadd. Apparently, the that two different days are two different epochs. In this case the "daily" and "epoch" coadds will be very very simi

- There are four distinct APOGEE spectra, and one coadd.

```
shape: (17,)
Series: 'allspec_id' [str]
[
  "sdss5-apo-apogee-1.3-4054-5965..."
  "sdss5-apo-apogee-1.3-4003-5965..."
  "sdss5-apo-apogee-1.3-4053-5965..."
  "sdss5-apo-apogee-1.3-4036-5965..."
  "sdss5-apo-apogee-1.3-2M0653379..."
  ...
  "sdss5-apo-boss-epoch-v6_2_1-10..."
  "sdss5-apo-boss-epoch-v6_2_1-10..."
  "sdss5-apo-boss-epoch-v6_2_1-10..."
  "sdss5-apo-boss-epoch-v6_2_1-10..."
  "sdss5-apo-boss-epoch-v6_2_1-10..."
]
```

Looking at the spectra

We can track down the spectra on disk quite easily. The `sas_url` tells us the path. We just need to change the root of

If the data don't already exist on disk (e.g., if you're not running this notebook on SciServer), we can download the dat

Here then are the paths in the local SAS directory structure:

```
/data/sdss/dr19/spectro/apogee/redux/1.3/visit/apo25m/103442/4054/59657/apVisit-1.3-apo25m-4054-59657-:
/data/sdss/dr19/spectro/apogee/redux/1.3/visit/apo25m/103442/4003/59654/apVisit-1.3-apo25m-4003-59654-:
/data/sdss/dr19/spectro/apogee/redux/1.3/visit/apo25m/103427/4053/59657/apVisit-1.3-apo25m-4053-59657-:
/data/sdss/dr19/spectro/apogee/redux/1.3/visit/apo25m/103442/4036/59655/apVisit-1.3-apo25m-4036-59655-:
/data/sdss/dr19/spectro/apogee/redux/1.3/stars/apo25m/100/100755/apStar-1.3-apo25m-2M06533798-0119171.1
/data/sdss/dr20/spectro/boss/redux/v6_2_1/spectra/daily/lite/103XXX/103427/59654/spec-103427-59654-270:
/data/sdss/dr20/spectro/boss/redux/v6_2_1/spectra/daily/lite/103XXX/103427/59655/spec-103427-59655-270:
/data/sdss/dr20/spectro/boss/redux/v6_2_1/spectra/daily/lite/103XXX/103427/59656/spec-103427-59656-270:
/data/sdss/dr20/spectro/boss/redux/v6_2_1/spectra/daily/lite/103XXX/103442/60657/spec-103442-60657-630!
/data/sdss/dr20/spectro/boss/redux/v6_2_1/spectra/daily/lite/103XXX/103442/60663/spec-103442-60663-630!
/data/sdss/dr20/spectro/boss/redux/v6_2_1/spectra/daily/lite/103XXX/103442/60693/spec-103442-60693-630!
/data/sdss/dr20/spectro/boss/redux/v6_2_1/spectra/epoch/lite/103XXX/103427/59654/spec-103427-59654-270:
/data/sdss/dr20/spectro/boss/redux/v6_2_1/spectra/epoch/lite/103XXX/103427/59655/spec-103427-59655-270:
```

```
/data/sdss/dr20/spectro/boss/redux/v6_2_1/spectra/epoch/lite/103XXX/103427/59656/spec-103427-59656-270!  
/data/sdss/dr20/spectro/boss/redux/v6_2_1/spectra/epoch/lite/103XXX/103442/60657/spec-103442-60657-630!  
/data/sdss/dr20/spectro/boss/redux/v6_2_1/spectra/epoch/lite/103XXX/103442/60663/spec-103442-60663-630!  
/data/sdss/dr20/spectro/boss/redux/v6_2_1/spectra/epoch/lite/103XXX/103442/60693/spec-103442-60693-630!
```

We can open one of the BOSS files up to see what it has in it. We'll first just look at what the HDUs are called.

```
Opened FITS spectrum file: /data/sdss/dr20/spectro/boss/redux/v6_2_1/spectra/daily/lite/103XXX/103427/!  
Total HDU count: 5
```

```
FLUX  
COADD  
SPALL  
ZALL  
ZLINE
```

It looks like "COADD" actually has the spectrum. This is a table, and the columns have the fluxes, wavelengths, masks, etc. https://data.sdss.org/datamodel/files/BOSS_SPECTRO_REDUX/RUN2D/spectra/PLATE4/spec.html

```
COADD table shape: (4648,)
```

```
Available column names in COADD:  
( 'FLUX', 'LOGLAM', 'IVAR', 'AND_MASK', 'OR_MASK', 'WDISP', 'SKY', 'MODEL', 'WRESL' )
```

But if we want to know the units of FLUX or LOGLAM we can check:

```
TUNIT1 (FLUX unit): 10^-17 ergs/s/cm^2/Angs  
TUNIT2 (LOGLAM unit): log10(Angs)
```

Now we can plot and label our plot:

